Case Study: National Real-Time Payments System

Context

A country is launching its **National Real-Time Payments System** to enable instant digital payments between citizens, businesses, and financial institutions. The platform aims to become the primary backbone for all peer-to-peer (P2P) and peer-to-merchant (P2M) transactions, replacing cash and traditional delayed banking transfers.

The system must operate at national scale, handle massive concurrency, and provide near-instant transaction processing with extremely high reliability and security.

The success of the platform could potentially lead to **international expansion**, enabling cross-border real-time payments with multiple partner countries and banking networks.

Users

- Total Registered Users: 150 million
- Daily Active Users: 40 million
- Peak Concurrent Users: 5 million
- Peak Transactions Per Second (TPS): 50,000+

User Types:

- Individual citizens
- Merchants (online & offline)
- Banks and financial institutions
- Regulators and auditors

Supported Platforms:

- Mobile Apps (iOS, Android)
- Web Applications
- Merchant POS systems
- Third-party integrations (via APIs)

Functional Requirements

Core Payment Features

- User onboarding with Customer verification (KYC)
- Link multiple bank accounts per user
- Real-time P2P payments
- Real-time P2M payments via QR codes
- Transaction history and receipts
- Transaction status (success, pending, failed)

Banking & Settlement

- Integration with all major banks
- Inter-bank settlement and reconciliation
- Support for scheduled settlements
- Dispute management and chargebacks

Merchant Capabilities

- Merchant registration and verification
- QR-based payments
- Merchant dashboards
- Daily settlement reports

Operational Features

- Admin dashboards
- Regulatory reporting
- Audit logs
- Fraud detection and risk scoring

Non-Functional Requirements

Performance & Scale

- End-to-end transaction latency < 2 seconds
- System availability: 99.999%
- Horizontal scalability for peak events
- Zero data loss

Security & Compliance

- PCI DSS compliance
- ISO 27001
- Data encryption (in transit & at rest)
- Strong authentication and authorization
- Compliance with national banking regulations
- Data residency and sovereignty laws

Reliability

- No single point of failure
- Automatic failover
- Graceful degradation
- Real-time monitoring and alerting

Additional Instructions / Business Constraints

1. The system must support **real-time fraud detection** without delaying transactions.
2. The platform should work even during **partial bank outages**.
3. During national events (festivals, salary days, tax deadlines), transaction volume can spike up to **10x**.
4. In future the system will be replicated for **international payments**, requiring:
    - Currency conversion
    - Cross-border settlement
    - Integration with foreign banking systems
    - Compliance with international financial regulations

Architectural Discussion Questions

Considering the above requirements, brainstorm and document your responses for the following:

1. What are the various systems and subsystems required to deliver this functionality?
2. What kind of information is exchanged between systems?
    - Consider frequency, speed, volume, and consistency requirements.

3. What are the major potential failure points in this system?
    - Technical failures
    - Integration failures
    - External dependency failures
4. What mitigation strategies would you design to prevent or reduce failures?
    - For Eg : Redundancy, Circuit breakers etc
5. What contingency plans are required when failures do occur?
    - Manual reconciliation
    - Deferred settlements
    - User communication strategies
6. How should the system behave differently under:
    - Normal daily load
    - Peak national events
7. What architectural trade-offs would you make between:
    - Consistency vs availability
    - Cost vs reliability
    - Real-time vs eventual processing
    - Security Depth vs User Experience
8. Refer to similar global implementations and highlight the USPs and additional features that your solution recommends. Highlight the impact and ROI of these additions.

Expected Outcomes for Participants

By the end of this case study, participants should be able to:

- Identify large-scale distributed system components
- Design for high availability and resilience
- Apply system thinking to real-world financial platforms
- Understand regulatory and operational constraints
- Justify architectural trade-offs under business pressure

This case study focuses on **system thinking, not technology choices**. Participants should reason at the architectural level rather than specific tools or vendors.